

Measurement-Decoupled Attention–Koopman–T–S Fuzzy Monitoring via Structured Basis-Perturbation Residual Learning

Anonymous Authors

Abstract—Monitoring nonlinear closed-loop systems without physical fault models, labeled fault samples, or known fault directions requires a structural solution to a fundamental difficulty: measurements that carry abnormal information also update the reference against which abnormality is judged. This paper addresses that difficulty through three connected constructions. First, a strictly causal attention encoder is coupled with a Takagi–Sugeno (T–S) fuzzy Koopman model to form a measurement-decoupled normal reference; a start-indexed residual then provides a single mathematical object serving free-rollout training, short-horizon monitoring, and long-reference monitoring. Second, the complete fuzzy Koopman Jacobian is decomposed into the interpolated local dynamics and the variation of normalized rule weights, yielding an implementable infinity-norm propagation bound and a reliable reference horizon without eigenvalue, singular-value, inverse-matrix, or higher-order derivative computations. Third, model-internal basis perturbations applied separately to input, measurement, and lifted coordinates generate finite-horizon residual responses through forward rollouts; the resulting responses are used to jointly reduce normal propagation, increase perturbation sensitivity, and separate response patterns. Online scores are standardized by fuzzy operating region, controller command, command variation, and reference age, with thresholds calibrated on held-out normal blocks. Sufficient residual-space conditions are derived for detection and unique group separation. When the conditions for a unique decision are not met, the method returns a set of compatible perturbation groups rather than an unsupported physical fault label.

Index Terms—Attention, fault detection, Koopman operator, normal-data-only learning, structured perturbation, Takagi–Sugeno fuzzy systems.

I. INTRODUCTION

NONLINEAR closed-loop systems commonly traverse multiple operating regimes while yielding few, if any, representative fault records. Learning normal input–output behavior is therefore an attractive monitoring strategy, yet it introduces a structural obstacle: the same measurements that form a residual are also used to update the state and model against which that residual is evaluated. When an abnormal measurement feeds an encoder, an attention context, or a fuzzy scheduling variable, the normal reference may drift toward the abnormal trajectory, reducing the very discrepancy that detection relies upon. This measurement-to-reference contamination creates a coupled design problem that cannot be resolved by any single module in isolation.

Koopman operator theory seeks lifted coordinates in which nonlinear dynamics admit a finite-dimensional linear approximation [1], [2]. T–S fuzzy models interpolate local

linear descriptions through normalized firing strengths, providing a principled framework for multi-regime dynamics [3]–[5]. Their combination has been investigated in modeling and predictive control [6], but these prior efforts target control-oriented prediction accuracy rather than the diagnostic requirements that arise when neither fault models nor fault data are available. The combination itself is consequently not treated as a contribution here. What is new is the operational role assigned to the T–S structure: it supplies the normal reference, it determines the finite-horizon error propagation bound through its complete Jacobian, and it conditions the online monitoring decision on the realized operating region. Attention is retained to represent long input–output histories compactly, but its output is prevented from routing new measurements into the normal reference after the initial anchor is established.

Classical fuzzy fault-detection methods introduce explicit fault or unknown-input channels and design observers or filters through Lyapunov and linear-matrix-inequality techniques [7]–[11]. Koopman-based fault diagnosis has similarly exploited learned residual and parity relations [12], [13]. These formulations are well suited when a physical fault model or representative fault data are available. In the setting considered here, however, training data are exclusively normal, and the physical fault location, direction, and temporal profile are all unknown. A physical minimum fault gain cannot therefore be identified; the only structure available for diagnosis consists of the known input and measurement coordinates together with the coordinates of the learned lifted model.

This restricted information boundary fuses three difficulties into one connected design problem. Excluding new measurements from the reference prevents it from adapting to abnormal observations, but simultaneously exposes the normal free-rollout drift that one-step teacher forcing would have masked. Reducing that drift alone is insufficient, because a less sensitive model may also suppress abnormal residuals. Moreover, an unknown physical fault direction precludes any justified low-dimensional fault-sensitive projection. The method must instead control normal propagation, retain the full joint residual, and learn separable responses solely for explicitly defined model-coordinate perturbation groups.

The principal contributions are as follows.

- 1) A measurement-decoupled attention–Koopman–T–S reference and a start-indexed residual are constructed. The residual object serves free-rollout training, short-horizon monitoring, and long-reference monitoring and yields an exact nonlinear residual recursion, while a

delay-confirmed anchor rule protects the reference from abnormal measurements.

- 2) The complete T-S fuzzy Koopman Jacobian is decomposed to expose the contributions of local dynamics, rule overlap, local-model disparity, and scheduling sensitivity. An infinity-norm bound, enforced by elementwise parameterization, supplies a finite-horizon normal residual bound and a reliable reference horizon without costly or poorly differentiable matrix operations.
- 3) Perturbation rollouts applied to the input, measurement, and lifted coordinates are integrated with normal-propagation learning. The resulting input-conditioned statistic admits sufficient residual-space detection and group-separation conditions, and returns a set-valued decision when a unique group is not supported.

The three contributions are interdependent by construction: the reference defined in the first contribution is the object whose propagation the second contribution bounds, and the perturbation responses learned in the third contribution operate on that same reference through ordinary forward rollouts. Section II formalizes the information boundary under which these constructions must operate and states the three subproblems in precise terms. Section III develops the normal model, measurement-decoupled reference, propagation analysis, and structured response learning. Section IV turns the frozen residual and learned patterns into an online detection and isolation decision rule. Section V summarizes the results and their scope.

Notation. Throughout the paper, the superscripts n and f identify quantities associated with normal and faulty operation, respectively; the subscript 0 is reserved for an initial value. Vectors and matrices are set in boldface; calligraphic capitals denote nonlinear mappings ($\mathcal{X}, \mathcal{Y}, \mathcal{T}, \mathcal{D}$), the encoder mapping $\mathcal{H}_\Theta$, the history set $\mathcal{H}_k^-$, and loss functionals $\mathcal{L}$; blackboard-bold capitals ($\mathbb{R}, \mathbb{G}, \mathbb{X}, \mathbb{Y}, \mathbb{Z}$) denote number fields, sets, and spaces. Dimension prefixes use m (e.g., m_u, m_x, m_y, m_z). The base families f/F and n are reserved for fault and normal quantities, respectively. The infinity norm $\|\cdot\|_\infty$ is the default norm unless otherwise stated; the ℓ_1 norm $\|\cdot\|_1$ is used for pattern distances and window scores. Standard basis vectors are denoted by $\mathbf{q}$ with a superscript identifying the coordinate space.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Information Boundary and Controlled-Process Description

Consider a closed-loop nonlinear system in which the controller issues a command $\mathbf{u}_k \in \mathbb{R}^{m_u}$ at each sample k . The actuator output $\mathbf{u}_k^a \in \mathbb{R}^{m_u}$ actually applied to the controlled process is not measured online, and may differ from the command because of actuator-side deviations. The fault-inclusive controlled process is described by

$$\begin{aligned} \mathbf{u}_k^a &= \mathbf{u}_k + \mathbf{f}_k^a, \\ \mathbf{x}_{k+1} &= \mathcal{X}^n(\mathbf{x}_k, \mathbf{u}_k^a) + \mathbf{f}_k^p + \boldsymbol{\varepsilon}_k^x, \\ \mathbf{y}_k &= \mathcal{Y}^n(\mathbf{x}_k, \mathbf{u}_k^a) + \mathbf{f}_k^s + \boldsymbol{\varepsilon}_k^y, \end{aligned} \quad (1)$$

where $\mathbf{x}_k \in \mathbb{R}^{m_x}$ and $\mathbf{y}_k \in \mathbb{R}^{m_y}$ denote the physical state and measured output, respectively. The additive vectors $\mathbf{f}_k^a \in \mathbb{R}^{m_u}$, $\mathbf{f}_k^p \in \mathbb{R}^{m_x}$, and $\mathbf{f}_k^s \in \mathbb{R}^{m_y}$ represent possible actuator-side, process-side, and measurement-side deviations from normal operation, while $\boldsymbol{\varepsilon}_k^x \in \mathbb{R}^{m_x}$ and $\boldsymbol{\varepsilon}_k^y \in \mathbb{R}^{m_y}$ denote the disturbances entering the state transition and the output measurement, respectively. The normal-operation mappings $\mathcal{X}^n : \mathbb{R}^{m_x} \times \mathbb{R}^{m_u} \rightarrow \mathbb{R}^{m_x}$ and $\mathcal{Y}^n : \mathbb{R}^{m_x} \times \mathbb{R}^{m_u} \rightarrow \mathbb{R}^{m_y}$ are unknown. The dimensions of the fault vectors specify only where a deviation can enter the physical description; they do not supply known fault directions to the learning algorithm.

Online fault diagnosis has access only to the recorded command $\mathbf{u}_k$ and the measured output $\mathbf{y}_k$. Conditioning the normal model on the recorded command answers a specific counterfactual question: how would the controlled process evolve under this command if its actuator and dynamics followed the normal-operation description? A command that a feedback controller modifies after sensing an abnormal measurement remains a known conditioning variable; the unmeasured discrepancy between $\mathbf{u}_k^a$ and $\mathbf{u}_k$ remains an inconsistency to be detected.

The learning regime is deliberately restricted. Only normal trajectories are used for model training and threshold calibration; no physical fault matrix, fault direction, fault derivative, fault time model, or labeled fault trajectory is employed. All model parameters and calibration quantities are frozen during testing.

B. Finite-History Lifted Representation

The information boundary defined above prohibits access to the physical state $\mathbf{x}_k$ and to the normal-operation mappings $\mathcal{X}^n, \mathcal{Y}^n$. A representation that can be constructed solely from the available history $\mathcal{H}_k^-$ is therefore required. Koopman operator theory provides the conceptual foundation for such a representation, but the ideal Koopman formulation must be adapted to what the information boundary permits.

For the normal-operation dynamics, the Koopman operator acts on an observable by composition with the state transition [1]. A finite vector of observables $\boldsymbol{\psi} : \mathbb{R}^{m_x} \rightarrow \mathbb{R}^{m_z}$ would satisfy the ideal relation

$$\boldsymbol{\psi}(\mathcal{X}^n(\mathbf{x}_k, \mathbf{u}_k)) = \boldsymbol{\psi}(\mathbf{x}_{k+1}). \quad (2)$$

In practice, a finite invariant subspace may not exist, and the learned lifted coordinates need not contain the physical state. The present formulation therefore adopts a predictive coordinate $\mathbf{z}_k^{\text{enc}} \in \mathbb{R}^{m_z}$ estimated from the finite history (3) and uses a T-S interpolation of local lifted linear models to approximate (2) on the normal operating domain. Both the approximation error inherent in finite lifting and the error due to finite history are retained in the subsequent propagation analysis.

C. Problem Formulation

Under the information boundary stated above, this paper addresses three connected subproblems.

- 1) *Reference construction*: construct a normal reference that admits new measurements for residual evaluation but not for reference updates, while using a single

residual definition across free-rollout training, short-horizon monitoring, and long-reference testing.

- 2) *Normal propagation*: quantify the finite-horizon drift of that reference through the complete attention–Koopman–T–S mapping using operations compatible with first-order network training, and make the realized input and fuzzy operating region explicit in the resulting bound.
- 3) *Residual-space decision*: using normal data alone, jointly reduce normal residual propagation and increase the separability of reproducible model-coordinate perturbation responses; then construct an input-conditioned detection statistic and return a unique group only when the residual response provides sufficient evidence.

III. PROPOSED METHOD

A. Attention–Koopman–T–S Normal Model

The normal-operation model is built from three components that share a common information discipline: every learned mapping accesses only the strictly past history and the recorded command $\mathbf{u}_k$. To enforce strict causality, the history available at sample k excludes the current output and is defined as

$$\mathcal{H}_k^- = \{\mathbf{u}_{k-\ell_h:k-1}, \mathbf{y}_{k-\ell_h:k-1}\}, \quad (3)$$

where ℓ_h is the history length.

Each element of the history is first embedded as a token $\mathbf{h}_j$. For attention head $q \in \{1, \dots, m_{\text{hd}}\}$, the strictly causal attention coefficient assigned at sample k to a past token j is computed as

$$\varpi_{k,j}^{(q)} = \frac{\exp\left((\mathbf{W}_Q^{(q)} \mathbf{h}_{k-1})^\top (\mathbf{W}_K^{(q)} \mathbf{h}_j) / \sqrt{m_a}\right)}{\sum_{\ell=k-\ell_h}^{k-1} \exp\left((\mathbf{W}_Q^{(q)} \mathbf{h}_{k-1})^\top (\mathbf{W}_K^{(q)} \mathbf{h}_\ell) / \sqrt{m_a}\right)}. \quad (4)$$

The query is formed from the most recent history token $\mathbf{h}_{k-1}$, ensuring that no information from sample k or later enters the attention weights. The head outputs are concatenated and projected to produce a compact context vector:

$$\begin{aligned} \chi_k &= \mathbf{W}_O \left[\sum_{j=k-\ell_h}^{k-1} \varpi_{k,j}^{(q)} \mathbf{W}_V^{(q)} \mathbf{h}_j \right]_{q=1}^{m_{\text{hd}}}, \\ \mathbf{z}_k^{\text{enc}} &= \mathcal{H}_\Theta(\mathcal{H}_k^-). \end{aligned} \quad (5)$$

The encoder $\mathcal{H}_\Theta$ encompasses the token embeddings and the multihead attention operation of (4). Attention here serves to compress the strictly past history; it is not treated as a separate contribution.

Having formed the encoded state and context, the model couples them with the controller command to produce a scheduling vector that drives the fuzzy rule memberships:

$$\boldsymbol{\rho}_k = \mathbf{W}_z \mathbf{z}_k + \mathbf{W}_\chi \chi_k + \mathbf{W}_u \mathbf{u}_k. \quad (6)$$

Constant offsets are absorbed through an augmented constant coordinate. For each rule $i \in \{1, \dots, m_r\}$, the negative squared Mahalanobis-like energy and the normalized firing strength are

$$\begin{aligned} \varphi_i(\boldsymbol{\rho}) &= -\frac{1}{2}(\boldsymbol{\rho} - \mathbf{c}_i)^\top \boldsymbol{\Pi}_i(\boldsymbol{\rho} - \mathbf{c}_i), \\ \omega_i(\boldsymbol{\rho}) &= \frac{\exp(\varphi_i(\boldsymbol{\rho}))}{\sum_{j=1}^{m_r} \exp(\varphi_j(\boldsymbol{\rho}))}. \end{aligned} \quad (7)$$

By construction, $\omega_i \geq 0$ for every rule and $\sum_i \omega_i = 1$, forming a valid convex combination. The diagonal metric $\boldsymbol{\Pi}_i$ and center $\mathbf{c}_i$ of each rule are parameterized with built-in bounds to guarantee positive definiteness without constrained optimization:

$$\begin{aligned} \boldsymbol{\Pi}_i &= \text{Diag}[\underline{\pi} \mathbf{1} + (\bar{\pi} - \underline{\pi}) \text{sigmoid}(\tilde{\boldsymbol{\pi}}_i)], \\ \mathbf{c}_i &= \bar{\mathbf{c}} \odot \tanh(\tilde{\mathbf{c}}_i). \end{aligned} \quad (8)$$

All operations inside the brackets are elementwise, and the constants satisfy $0 < \underline{\pi} < \bar{\pi}$. Although $\boldsymbol{\Pi}_i$ is diagonal, cross-coordinate dependence enters through the linear map (6); a diagonal metric in the scheduling space does not imply an axis-aligned partition of the original history space.

The local and interpolated lifted dynamics and the output decoder are defined as

$$\begin{aligned} \mathbf{v}_i(\mathbf{z}, \mathbf{u}) &= \mathbf{A}_i \mathbf{z} + \mathbf{B}_i \mathbf{u}, \\ \mathcal{T}_\Theta(\mathbf{z}, \mathbf{u}; \boldsymbol{\chi}) &= \sum_{i=1}^{m_r} \omega_i(\boldsymbol{\rho}) \mathbf{v}_i(\mathbf{z}, \mathbf{u}), \\ \mathbf{y}^{\text{dec}} &= \mathcal{D}_\Theta(\mathbf{z}, \mathbf{u}). \end{aligned} \quad (9)$$

The decoder $\mathcal{D}_\Theta$ is distinct from the physical output mapping $\mathcal{Y}^n$; this separation is necessary because the learned lifted state does not necessarily contain the physical state.

To obtain a computable propagation bound without spectral decomposition, the matrices that govern the bound are parameterized elementwise through a bounded hyperbolic-tangent map, which enforces the required bounds by construction. For example,

$$\begin{aligned} [\mathbf{A}_i]_{pq} &= \frac{\kappa_A}{m_z} \tanh([\tilde{\mathbf{A}}_i]_{pq}), \\ [\mathbf{B}_i]_{pq} &= \frac{\kappa_B}{m_u} \tanh([\tilde{\mathbf{B}}_i]_{pq}), \\ [\mathbf{W}_z]_{pq} &= \frac{\kappa_W}{m_z} \tanh([\tilde{\mathbf{W}}_z]_{pq}). \end{aligned} \quad (10)$$

Consequently, $\|\mathbf{A}_i\|_\infty \leq \kappa_A$, $\|\mathbf{B}_i\|_\infty \leq \kappa_B$, and $\|\mathbf{W}_z\|_\infty \leq \kappa_W$ hold by construction, without requiring spectral projection or post-training verification.

Training uses a normal data split and a free-rollout horizon ℓ_{tr} . Starting from the encoded state at a sampled start s , i.e., $\mathbf{z}_{s|s}^{\text{roll}} = \mathbf{z}_s^{\text{enc}}$, the model propagates forward, updating the history from its own predicted outputs rather than from the true measurements; here $\mathcal{H}_\Theta^\chi$ denotes the context-producing component of the encoder $\mathcal{H}_\Theta$, and $\mathcal{H}_{t|s}^{\text{roll}}$ is the rollout history assembled from self-predicted outputs:

$$\begin{aligned} \chi_{t|s}^{\text{roll}} &= \mathcal{H}_\Theta^\chi(\mathcal{H}_{t|s}^{\text{roll}}), \\ \mathbf{z}_{t+1|s}^{\text{roll}} &= \mathcal{T}_\Theta(\mathbf{z}_{t|s}^{\text{roll}}, \mathbf{u}_t; \chi_{t|s}^{\text{roll}}), \\ \mathbf{y}_{t|s}^{\text{roll}} &= \mathcal{D}_\Theta(\mathbf{z}_{t|s}^{\text{roll}}, \mathbf{u}_t). \end{aligned} \quad (11)$$

Free rollout, rather than one-step teacher forcing, is essential because the online measurement-decoupled reference, once initialized from an accepted anchor, propagates without access to future measurements. The normal prediction loss is therefore formulated directly on the free-rollout trajectory:

$$\mathcal{L}_{\text{pred}} = \sum_s \sum_{\ell=1}^{\ell_{\text{tr}}} (\lambda_z \left\| \mathbf{z}_{s+\ell}^{\text{enc}} - \mathbf{z}_{s+\ell}^{\text{roll}} \right\|_1 + \lambda_y \left\| \mathbf{y}_{s+\ell} - \mathbf{y}_{s+\ell}^{\text{roll}} \right\|_1). \quad (12)$$

The ℓ_1 norm is chosen over the more common ℓ_2 norm because it produces a tighter elementwise propagation bound in Corollary 1 and aligns with the ℓ_1 pattern distances used later for isolation. The two loss weights λ_z and λ_y control the trade-off between latent-state accuracy and output reconstruction fidelity.

Figure 1 makes this separation explicit. The data branch continuously encodes the strictly past history and produces the encoded state $\mathbf{z}_k^{\text{enc}}$, which enters the residual directly. The measurement-decoupled reference branch propagates $\mathbf{z}_k^n$ from the latest delay-confirmed normal anchor using only the frozen model and the recorded command; no new measurement corrects this reference after the anchor is accepted. The joint residual $\mathbf{e}_k$ compares the encoded state against both a short rolling reference and the long measurement-decoupled reference. Structured basis perturbations are injected into the frozen model to generate diagnostic responses, and the online decision module produces a score, a dynamic threshold, and either a unique group or a compatible set.

B. Measurement-Decoupled Reference and Unified Residual

A single residual object serves training, short-horizon monitoring, and long-reference monitoring, differing only in the choice of the rollout start s . For any start index s , the rollout (11) defines the lifted residual

$$\mathbf{e}_{k|s}^z = \mathbf{z}_k^{\text{enc}} - \mathbf{z}_{k|s}^{\text{roll}}, \quad k \geq s, \quad (13)$$

and the instantaneous output-consistency residual is

$$\mathbf{e}_k^y = \mathbf{y}_k - \mathcal{D}_{\Theta}(\mathbf{z}_k^{\text{enc}}, \mathbf{u}_k). \quad (14)$$

The same lifted residual (13) appears in the training loss (12); online, only the start index changes. Setting $s = k - \ell_{\text{sh}}$ gives a short rolling residual over the most recent ℓ_{sh} samples; setting $s = s_k^n$, where s_k^n is the latest delay-confirmed normal anchor (the confirmation procedure is defined below), gives the long-reference residual. The long reference state and its age are defined as

$$\mathbf{z}_k^n = \mathbf{z}_{k|s_k^n}^{\text{roll}}, \quad \ell_k^n = k - s_k^n. \quad (15)$$

The joint residual stacks all three components:

$$\mathbf{e}_k = \begin{bmatrix} (\mathbf{e}_k^y)^{\top} & (\mathbf{e}_{k|k-\ell_{\text{sh}}}^z)^{\top} & (\mathbf{e}_{k|s_k^n}^z)^{\top} \end{bmatrix}^{\top}. \quad (16)$$

Anchor management follows a delay-confirmation discipline. An encoded state is not immediately accepted as a normal anchor. It is first stored as a candidate and committed only after ℓ_{cf} consecutive warning-free samples elapse. During warning and alarm intervals, the accepted anchor is frozen and (11)

continues without measurement updates, preventing abnormal measurements from contaminating the reference.

Proposition 1 (Delayed anchor acceptance). *Assume that every abnormal trajectory belonging to a specified detectable class triggers a warning within at most ℓ_{det} samples of its onset. If $\ell_{\text{cf}} \geq \ell_{\text{det}}$, no candidate generated at or after that onset is committed before the warning.*

Proof. Commitment of a candidate requires ℓ_{cf} consecutive warning-free samples. Under the stated premise, a warning occurs no later than $\ell_{\text{det}} \leq \ell_{\text{cf}}$ samples after onset, so any post-onset candidate fails the acceptance condition. $\square$

Proposition 1 is therefore a conditional guarantee: faults that produce no detectable change in the available input–output information cannot be covered. The discipline is re-applied after every alarm: the method waits at least ℓ_h samples so that the encoder history no longer spans the flagged interval, and then repeats the delay-confirmation procedure.

To characterize how the start-indexed residual evolves, two intermediate quantities that isolate distinct sources of model mismatch are introduced:

$$\begin{aligned} \boldsymbol{\nu}_k &= \mathbf{z}_{k+1}^{\text{enc}} - \mathcal{T}_{\Theta}(\mathbf{z}_k^{\text{enc}}, \mathbf{u}_k; \boldsymbol{\chi}_k), \\ \boldsymbol{\delta}_{k|s}^{\chi} &= \mathcal{T}_{\Theta}(\mathbf{z}_k^{\text{enc}}, \mathbf{u}_k; \boldsymbol{\chi}_k) - \mathcal{T}_{\Theta}(\mathbf{z}_k^{\text{enc}}, \mathbf{u}_k; \boldsymbol{\chi}_{k|s}^{\text{roll}}). \end{aligned} \quad (17)$$

The one-step transition residual $\boldsymbol{\nu}_k$ captures the one-step inconsistency of the encoded state with the transition model evaluated at the encoding context; the history feature substitution error $\boldsymbol{\delta}_{k|s}^{\chi}$ measures the effect of replacing the encoding context $\boldsymbol{\chi}_k$ (which benefits from the true measurement history) with the rollout context $\boldsymbol{\chi}_{k|s}^{\text{roll}}$ (which uses only self-predicted outputs).

Theorem 1 (Exact start-indexed residual propagation). *Assume $\mathcal{T}_{\Theta}$ is continuously differentiable in $\mathbf{z}$ on the line segment joining $\mathbf{z}_{k|s}^{\text{roll}}$ and $\mathbf{z}_k^{\text{enc}}$. Then the lifted residual obeys the exact one-step recursion*

$$\mathbf{e}_{k+1|s}^z = \bar{\Gamma}_{k|s} \mathbf{e}_{k|s}^z + \boldsymbol{\nu}_k + \boldsymbol{\delta}_{k|s}^{\chi}, \quad (18)$$

where the averaged Jacobian is

$$\bar{\Gamma}_{k|s} = \int_0^1 \Gamma_{z,k}(\mathbf{z}_{k|s}^{\text{roll}} + \tau \mathbf{e}_{k|s}^z) \, d\tau. \quad (19)$$

Consequently, the finite-horizon propagation admits the explicit expansion

$$\mathbf{e}_{k|s}^z = \Phi(k, s) \mathbf{e}_{s|s}^z + \sum_{j=s}^{k-1} \Phi(k, j+1) (\boldsymbol{\nu}_j + \boldsymbol{\delta}_{j|s}^{\chi}), \quad (20)$$

with the ordered product $\Phi(k, j) = \bar{\Gamma}_{k-1|s} \cdots \bar{\Gamma}_{j|s}$ and the boundary condition $\Phi(j, j) = \mathbf{I}$.

The proof proceeds by adding and subtracting the transition evaluated at the encoding context $\boldsymbol{\chi}_k$ and at the rollout context $\boldsymbol{\chi}_{k|s}^{\text{roll}}$, both at the encoded state $\mathbf{z}_k^{\text{enc}}$, followed by the integral mean-value identity; the complete derivation is given in Appendix A. The recursion consequently exposes the two sources of finite-horizon forcing: the one-step transition residual $\boldsymbol{\nu}_k$ inherent in the finite-history encoding, and the history feature substitution error $\boldsymbol{\delta}_{k|s}^{\chi}$ that accumulates with the attention memory length.

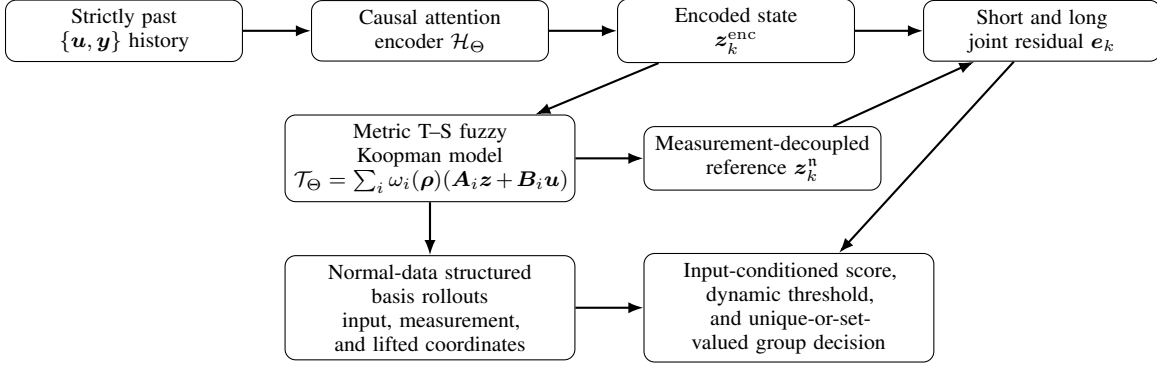


Fig. 1. Information paths of the proposed method. The architecture separates two roles: the data branch (top row) reads new measurements and feeds them into the residual, while the measurement-decoupled reference branch (lower left) is propagated by the frozen model using only the recorded command and self-predicted outputs. Structured basis rollouts (lower center) inject model-coordinate perturbations and compare the resulting responses against the unperturbed normal branch to learn separable patterns.

C. Fuzzy-Structure-Dependent Normal Propagation

Theorem 1 reduces the residual evolution to the product of averaged Jacobians and the accumulated forcing. Obtaining a computable bound on the Jacobian requires analyzing how the complete T-S fuzzy structure—not merely the weighted local matrices—responds to changes in the lifted state.

For compactness, define the local next-state value $\mathbf{v}_{i,k} = \mathbf{A}_i \mathbf{z}_k + \mathbf{B}_i \mathbf{u}_k$ and its membership-weighted average $\bar{\mathbf{v}}_k = \sum_i \omega_{i,k} \mathbf{v}_{i,k}$. From the membership definition (7), the gradient of each rule’s metric energy with respect to the scheduling vector is

$$\begin{aligned} \gamma_{i,k}^\rho &:= \nabla_\rho \varphi_{i,k} = -\mathbf{\Pi}_i(\rho_k - \mathbf{c}_i), \\ \bar{\gamma}_k^\rho &:= \sum_j \omega_{j,k} \gamma_{j,k}^\rho, \\ \nabla_\rho \omega_{i,k} &= \omega_{i,k} (\gamma_{i,k}^\rho - \bar{\gamma}_k^\rho). \end{aligned} \quad (21)$$

Here $\gamma_{i,k}^\rho$ (lowercase, with superscript ρ) denotes the local energy gradient for rule i , while $\mathbf{\Gamma}_{z,k}$ (uppercase, capital gamma, with subscript z) will denote the complete Jacobian of the interpolated transition with respect to the lifted state; the notational distinction reflects their differing roles in the fuzzy structure. The difference $\gamma_{i,k}^\rho - \bar{\gamma}_k^\rho$ quantifies how strongly rule i ’s local metric gradient deviates from the membership-weighted average. This term is small in regions where a single rule dominates and becomes significant in transition zones where multiple rules compete.

Theorem 2 (Complete fuzzy lifted-state Jacobian). *For fixed $\mathbf{u}_k$ and χ_k , the Jacobian of the interpolated transition (9) with respect to $\mathbf{z}$ decomposes as*

$$\begin{aligned} \mathbf{\Gamma}_{z,k} &= \sum_{i=1}^{m_r} \omega_{i,k} \mathbf{A}_i \\ &+ \sum_{i=1}^{m_r} \omega_{i,k} (\mathbf{v}_{i,k} - \bar{\mathbf{v}}_k) (\gamma_{i,k}^\rho - \bar{\gamma}_k^\rho)^\top \mathbf{W}_z. \end{aligned} \quad (22)$$

Moreover, its infinity norm satisfies the implementable bound

$$\begin{aligned} \|\mathbf{\Gamma}_{z,k}\|_\infty &\leq \kappa_k := \sum_i \omega_{i,k} \|\mathbf{A}_i\|_\infty \\ &+ \|\mathbf{W}_z\|_\infty \sum_i \omega_{i,k} \|\mathbf{v}_{i,k} - \bar{\mathbf{v}}_k\|_\infty \\ &\quad \times \left\| \gamma_{i,k}^\rho - \bar{\gamma}_k^\rho \right\|_1. \end{aligned} \quad (23)$$

Equation (22) separates two contributions that arise from distinct structural mechanisms of the fuzzy interpolation. The first sum is the direct interpolation of the local dynamics matrices—the term that would appear if membership weights were fixed. The second sum captures the variation of the normalized firing strengths with respect to the lifted state. This term becomes large only when three conditions coincide: multiple rules have non-negligible weights (rule overlap), their local next-state predictions differ substantially, and the scheduling vector lies in a direction that redistributes weight among those rules. The proof of the decomposition and the derivation of the induced-norm bound are given in Appendix B.

Equation (23) supplies the structural component of the propagation bound, while the unresolved remainder—arising from finite-data approximation, unmodeled disturbances, and encoder errors—is represented by an empirically calibrated forcing term.

Assumption 1 (Normal residual forcing). *On a prescribed normal operating domain, known nonnegative values $\bar{\epsilon}_k$ satisfy*

$$\left\| \boldsymbol{\nu}_k + \boldsymbol{\delta}_{k|s}^\chi \right\|_\infty \leq \bar{\epsilon}_k \quad (24)$$

on the calibrated normal event.

The values in Assumption 1 are not claimed as deterministic global bounds derived from finite data. The structural contribution comes from (10)–(23); the unresolved normal remainder is calibrated on held-out normal blocks in Section IV.

Corollary 1 (Finite-horizon normal residual bound). *If $\beta_{s|s} \geq \|e_{s|s}^z\|_\infty$ and the scalar sequence satisfies the linear recursion*

$$\beta_{k+1|s} = \kappa_k \beta_{k|s} + \bar{\epsilon}_k, \quad (25)$$

then $\|e_{k|s}^z\|_\infty \leq \beta_{k|s}$ holds while the trajectory remains in the prescribed normal domain. For a given reference tolerance $\bar{\beta}$, the reliable horizon is

$$\ell_{\text{ref}}(s) = \max \{ \ell : \beta_{s+j|s} \leq \bar{\beta}, 1 \leq j \leq \ell \}. \quad (26)$$

All quantities in (23) involve only matrix products, elementwise absolute values, and row or column sums—operations that are efficient and compatible with first-order automatic differentiation. The empirical training counterpart of (25),

$$\mathcal{L}_{\text{nr}} = \sum_s \sum_{\ell=1}^{\ell_{\text{tr}}} \beta_{s+\ell|s}^{\text{emp}}, \quad (27)$$

serves as a normal-reference propagation regularizer. By itself, it does not constitute a claim of improved physical fault sensitivity; its role is to control the normal side of residual separation.

D. Structured Basis-Perturbation Residual Learning

Reducing the normal propagation loss (27) controls only one side of the diagnostic problem. The regularizer (25) penalizes large values of the propagated bound $\beta_{k|s}$, but this bound enters the training objective multiplicatively with the Jacobian norm κ_k and additively with the forcing bound $\bar{\varepsilon}_k$. Because the same matrices A_i , B_i , and W_z that determine κ_k also determine the model's sensitivity to input and state perturbations, a parameter configuration that yields a small $\beta_{k|s}$ may simultaneously produce a small response to any perturbation—normal or abnormal. This is the well-known tension between nominal robustness and fault sensitivity in residual-based diagnosis: any mechanism that attenuates normal propagation also risks attenuating the abnormal response unless a separate diagnostic objective counterbalances it. Without physical fault data or a known fault direction, a diagnostic training signal must therefore be generated from structure that is explicitly available within the learned model.

To this end, define the model-coordinate perturbation group set

$$\mathbb{G}_b = \{u, y, z\}, \quad (28)$$

where u , y , and z identify the input, measurement, and lifted-coordinate branches, respectively. These labels specify where a perturbation is inserted in the learned architecture; they are not physical actuator, sensor, or process categories.

At a sampled start s , one standard basis vector and a signed amplitude α are selected. The perturbation is injected into exactly one branch while the rest of the model remains frozen:

- The input branch replaces u_s by $u_s + \alpha q_j^u$ in the model rollout, simulating a perturbation applied to the recorded command.
- The measurement branch replaces y_s by $y_s + \alpha q_j^y$ in the current output residual and in all subsequent causal histories that include this sample.
- The lifted branch adds the perturbation directly to the one-step transition:

$$z_{s+1}^{(z,j,\alpha)} = \mathcal{T}_\Theta(z_s^{\text{enc}}, u_s; \chi_s) + \alpha q_j^z. \quad (29)$$

After the injection, every branch follows the same frozen model and reference logic as the unperturbed normal branch. Denote the unperturbed joint residual at sample t by e_t^n .

For a response horizon ℓ_b , the perturbed and normal residual stacks are differenced to isolate the perturbation response:

$$\Delta e_{s,\ell_b}^{g,j,\alpha} = \begin{bmatrix} (e_s^{g,j,\alpha} - e_s^n)^\top & \cdots & (e_{s+\ell_b-1}^{g,j,\alpha} - e_{s+\ell_b-1}^n)^\top \end{bmatrix}^\top. \quad (30)$$

The amplitude-normalized sensitivity and the unit- ℓ_1 response pattern are defined as

$$\eta_{g,j,\alpha}^{\text{rsp}} = \frac{\|\Delta e_{s,\ell_b}^{g,j,\alpha}\|_1}{|\alpha| + \epsilon_{\text{reg}}}, \quad p_{g,j,\alpha} = \frac{|\Delta e_{s,\ell_b}^{g,j,\alpha}|}{\|\Delta e_{s,\ell_b}^{g,j,\alpha}\|_1 + \epsilon_{\text{reg}}}, \quad (31)$$

where the absolute value in the numerator of the pattern is elementwise and $\epsilon_{\text{reg}} > 0$ provides numerical stabilization.

With these quantities, the diagnostic side of training pursues two objectives that operate jointly with the normal-propagation regularizer: the sensitivity loss, which encourages each group's perturbation to produce a response of at least a target magnitude, and the separation loss, which enforces that response patterns from different groups remain distinguishable in ℓ_1 distance by at least a margin η_{iso} :

$$\begin{aligned} \mathcal{L}_{\text{sen}} &= \sum_{g,j,\alpha} \left[\eta_g - \eta_{g,j,\alpha}^{\text{rsp}} \right]_+^2, \\ \mathcal{L}_{\text{iso}} &= \sum_{g \neq \bar{g}} \left[\eta_{\text{iso}} - \|p_{g,j,\alpha} - p_{\bar{g},\bar{j},\bar{\alpha}}\|_1 \right]_+^2. \end{aligned} \quad (32)$$

The complete training objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{nr}} \mathcal{L}_{\text{nr}} + \lambda_{\text{sen}} \mathcal{L}_{\text{sen}} + \lambda_{\text{iso}} \mathcal{L}_{\text{iso}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}}, \quad (33)$$

where $\mathcal{L}_{\text{bal}}$ penalizes only rule occupancies that fall below a prescribed minimum; it does not enforce uniform rule usage. Each mini-batch samples one coordinate from each group, so the additional computation consists of three ordinary forward rollouts and first-order backpropagation. No Hessian, eigendecomposition, singular-value decomposition, matrix inverse, or online optimization is introduced by this construction.

The additive form of (33) mirrors the modular design of the method: each loss targets one requirement of the monitoring objective, and the weights λ_{nr} , λ_{sen} , λ_{iso} , and λ_{bal} govern the relative emphasis among the prediction, propagation, and diagnostic objectives, with larger values placing stronger weight on the corresponding property.

Having constructed the measurement-decoupled reference, the normal propagation bound, and the structured perturbation responses, we now formulate the online procedure that consumes these frozen objects to produce detection alarms and isolation decisions.

IV. INPUT-CONDITIONED DETECTION AND SET-VALUED ISOLATION

A. Input-Conditioned Residual Standardization

The raw joint residual (16) follows a distribution that shifts with the realized command, the command variation, the active

fuzzy operating region, and the age of the measurement-decoupled reference. Applying a fixed threshold across these changing conditions would confound normal operating-point variation with abnormality. The following standardization absorbs these dependencies into a condition-dependent scale before a scalar score is formed.

The condition descriptor collects the quantities that govern the residual distribution:

$$\mathbf{v}_k = \mathbf{u}_k - \mathbf{u}_{k-1}, \quad \boldsymbol{\zeta}_k = [(\boldsymbol{\omega}_k^n)^\top \quad \mathbf{u}_k^\top \quad \mathbf{v}_k^\top \quad \ell_k^n]^\top, \quad (34)$$

where $\boldsymbol{\omega}_k^n$ contains the firing strengths evaluated along the normal reference $\mathbf{z}_k^n$ and its predicted context. For each fuzzy rule, normal training blocks supply a local conditional mean $\boldsymbol{\mu}_i(\boldsymbol{\zeta}_k)$ and a positive elementwise scale $\sigma_i(\boldsymbol{\zeta}_k)$. The mixture moments incorporate both the within-rule uncertainty and the between-rule dispersion:

$$\begin{aligned} \boldsymbol{\mu}_k &= \sum_i \omega_{i,k}^n \boldsymbol{\mu}_i, \\ \boldsymbol{\sigma}_k^{\odot 2} &= \sum_i \omega_{i,k}^n [\boldsymbol{\sigma}_i^{\odot 2} + (\boldsymbol{\mu}_i - \boldsymbol{\mu}_k)^{\odot 2}] + \epsilon_\sigma \mathbf{1}. \end{aligned} \quad (35)$$

The between-rule term $(\boldsymbol{\mu}_i - \boldsymbol{\mu}_k)^{\odot 2}$ prevents the conditional variance from being underestimated at rule boundaries, where the residual mean can shift appreciably between adjacent operating regimes, and the additive floor $\epsilon_\sigma \mathbf{1}$ keeps every element of $\boldsymbol{\sigma}_k^{\odot 2}$ positive. Let σ_k denote the elementwise square root of $\boldsymbol{\sigma}_k^{\odot 2}$ in (35); elementwise standardization yields

$$\tilde{\mathbf{e}}_k = (\mathbf{e}_k - \boldsymbol{\mu}_k) \oslash \sigma_k. \quad (36)$$

For each window length ℓ drawn from a predeclared finite list $\{\ell_1, \dots, \ell_{\max}\}$, the accumulated ℓ_1 score is

$$s_{k,\ell} = \sum_{j=k-\ell+1}^k \|\tilde{\mathbf{e}}_j\|_1. \quad (37)$$

A held-out normal calibration split is partitioned by predeclared fuzzy operating cells and reference-age intervals. Within each cell, nonoverlapping blocks calibrate the upper quantile $\tau_{\text{cal}}(\boldsymbol{\zeta}_k, \ell)$. The propagated deterministic component from (25), after the same elementwise scaling, contributes $\tau_{\text{prop}}(\beta_{k|s_k^n}, \ell)$. The online threshold combines both sources:

$$\tau_{k,\ell} = \tau_{\text{cal}}(\boldsymbol{\zeta}_k, \ell) + \tau_{\text{prop}}(\beta_{k|s_k^n}, \ell). \quad (38)$$

The calibration term absorbs the unresolved random remainder; the propagation term tracks the accumulating reference drift. Together they produce a threshold that adapts to the input, input variation, fuzzy region, and reference age without online re-estimation. In practice, the two components play complementary roles: τ_{cal} dominates shortly after anchor acceptance, when the reference drift is small and the residual distribution is determined primarily by normal modeling error; τ_{prop} grows with reference age ℓ_k^n and eventually overtakes the calibration term, signaling that the reference has accumulated enough drift to warrant reconfirmation.

The multiscale statistic takes the maximum normalized score over the declared window family:

$$s_k^{\max} = \max_{\ell \in \{\ell_1, \dots, \ell_{\max}\}} \frac{s_{k,\ell}}{\tau_{k,\ell}}. \quad (39)$$

Its final decision level is calibrated directly on the maximum score, avoiding the multiplicity correction that separate per-window tests would require. Under serial dependence, the finite-sample coverage interpretation is blockwise rather than samplewise, conforming to the exchangeable-block calibration framework [14], [15].

B. Sufficient Detection Condition

The normal-propagation regularizer (27) and the structured response losses (32) act on opposite sides of the same residual. The following proposition makes explicit why both are necessary and how they combine to produce a detectable margin.

Proposition 2 (Residual-space detection). *For a selected window, suppose a normal standardized residual block satisfies $\|\tilde{\mathbf{e}}^n\|_1 \leq \tau_{k,\ell}$ on the calibrated normal event. Let the corresponding faulty residual block be $\tilde{\mathbf{e}}^f = \tilde{\mathbf{e}}^n + \Delta \tilde{\mathbf{e}}^f$, and suppose a structured perturbation produces the learned response $\Delta \tilde{\mathbf{e}}^{g,j,\alpha}$ with $\|\Delta \tilde{\mathbf{e}}^f - \Delta \tilde{\mathbf{e}}^{g,j,\alpha}\|_1 \leq \epsilon_g$. If*

$$\|\Delta \tilde{\mathbf{e}}^{g,j,\alpha}\|_1 > 2\tau_{k,\ell} + \epsilon_g, \quad (40)$$

then the corresponding residual block exceeds $\tau_{k,\ell}$ on that event.

The proof follows from the reverse triangle inequality: $\|\tilde{\mathbf{e}}^f\|_1 \geq \|\Delta \tilde{\mathbf{e}}^{g,j,\alpha}\|_1 - \epsilon_g - \tau_{k,\ell} > \tau_{k,\ell}$. Assuming the trained responses satisfy $\|\Delta \tilde{\mathbf{e}}^{g,j,\alpha}\|_1 \geq \eta_g |\alpha|$ after the standardization of (36), the condition translates into a sufficient amplitude requirement:

$$|\alpha| > \frac{2\tau_{k,\ell} + \epsilon_g}{\eta_g}. \quad (41)$$

This is a residual-model conclusion, not a physical minimum-fault claim. It exposes the two factors that the training objective can control: reducing normal propagation lowers the threshold $\tau_{k,\ell}$, and increasing the structured perturbation response raises the sensitivity η_g .

C. Unique-or-Set-Valued Isolation

After a detection is declared, the observed residual increment is normalized to a unit- ℓ_1 pattern by the same procedure as (31). For each group g , let $\delta_g(k)$ be the smallest ℓ_1 distance from the observed pattern to the patterns stored during training rollouts for that group. The compatible set collects every group whose distance falls within a calibrated tolerance:

$$\mathbb{G}_k^{\text{cmp}} = \{g \in \mathbb{G}_b : \delta_g(k) \leq \tau_{\text{iso}}\}. \quad (42)$$

A unique group is returned only when $\mathbb{G}_k^{\text{cmp}}$ contains exactly one element and the distance to the second-closest group exceeds a calibrated ambiguity margin. When neither condition holds, the observed residual pattern does not uniquely implicate one insertion site, and the method reports the full compatible set.

Proposition 3 (Sufficient group separation). *Suppose the observed normalized response lies within δ_{obs} of a stored pattern generated from the true insertion group. If every stored*

pattern belonging to a different group is at an ℓ_1 distance greater than $2\delta_{\text{obs}}$ from that stored pattern, nearest-pattern classification recovers the generating group.

The result follows from the triangle inequality. When its premise is unsatisfied—either because the observed pattern falls near a group boundary or because the perturbation responses of two groups overlap in the residual space—the compatible set (42) is returned. The labels u , y , and z indicate compatibility with a perturbation insertion site in the learned architecture. Without an independently verified mapping from model coordinates to physical components, they must not be reinterpreted as unique actuator, sensor, or process fault identifications.

D. Offline and Online Algorithms

The offline procedure, summarized in Algorithm 1, consumes disjoint normal training and calibration blocks. All learned parameters, statistics, thresholds, and response patterns are frozen before online deployment.

Algorithm 1 Normal-Data Offline Design

Require: Disjoint normal training and calibration blocks

- 1: Train $\mathcal{H}_\Theta$, $\mathcal{T}_\Theta$, and $\mathcal{D}_\Theta$ by free rollouts using (12).
 - 2: Apply (8) and (10); compute (23) and (27).
 - 3: Sample one coordinate from each group in (28); optimize (33) with ordinary forward branches.
 - 4: Freeze all model and fuzzy-rule parameters.
 - 5: Estimate (35) and calibrate (38) on held-out nonoverlapping normal blocks.
 - 6: Store response patterns and calibrate the ambiguity margin for (42).
-

Online, the encoder continuously reads the strictly past history while the measurement-decoupled reference either advances from its accepted anchor or awaits delayed reconfirmation. Algorithm 2 summarizes the per-sample decision loop.

Algorithm 2 Online Monitoring

Require: Frozen model, normal statistics, thresholds, and response patterns

- 1: **for** each sample k **do**
 - 2: Compute z_k^{enc} and e_k^y from the data branch.
 - 3: Propagate the short rollout and the measurement-decoupled reference.
 - 4: Form e_k by (16) and ζ_k by (34).
 - 5: Compute (36)–(39).
 - 6: **if** ζ_k^{max} exceeds its calibrated decision level **then**
 - 7: Freeze anchor acceptance and report an alarm.
 - 8: Return $\mathbb{G}_k^{\text{cmp}}$ by (42).
 - 9: **else**
 - 10: Continue delay-confirmed anchor management.
 - 11: **end if**
 - 12: After an alarm, wait ℓ_h samples and repeat delay confirmation before accepting a new anchor.
 - 13: **end for**
-

The online stage performs only encoder and model forward passes, elementwise standardization, vector norms, and the lookup-table threshold and score evaluations of (38) and (39). Fuzzy centers, metrics, network weights, and thresholds remain frozen; no online update, reclustering, matrix decomposition, or optimization is required.

V. CONCLUSION

This paper formulated normal-data-only nonlinear monitoring as three aligned problems whose solutions are structurally interdependent. Preventing new measurements from updating the normal reference, controlling the free-rollout drift that this decoupling exposes, and making residual-space decisions without physical fault priors cannot be addressed in isolation—each solution constrains and enables the next.

A measurement-decoupled attention–Koopman–T–S reference and a start-indexed residual were constructed so that one and the same residual object serves free-rollout training, short-horizon monitoring, and long-reference evaluation, avoiding the discrepancy that would arise if training and online operation used different residual definitions. The complete fuzzy Jacobian decomposition then linked normal propagation to four structurally distinct factors—local dynamics, rule overlap, local-model disparity, and scheduling sensitivity—and supplied an implementable infinity-norm bound that requires only elementwise operations and row or column sums. Structured model-coordinate perturbations, injected separately into the input, measurement, and lifted branches, complemented the normal-propagation objective by providing a diagnostic training signal that increases residual sensitivity and intergroup separation without physical fault data.

On the decision side, input-, fuzzy-region-, and reference-age-conditioned standardization produced a dynamic detection rule, with sufficient residual-space conditions that expose the complementary roles of propagation reduction and response amplification. The isolation output is deliberately conservative: when the observed residual pattern does not uniquely implicate one insertion location, the method returns a compatible set rather than an unsupported singleton.

The method does not claim unique physical fault identification: such a claim requires additional actuation measurements, fault-channel knowledge, or a verified mapping from lifted coordinates to physical components.

APPENDIX A

PROOF OF THEOREM 1

Start from the definition (13) of the lifted residual:

$$e_{k+1|s}^z = z_{k+1}^{\text{enc}} - z_{k+1|s}^{\text{roll}}.$$

Add and subtract the transition $\mathcal{T}_\Theta(z_k^{\text{enc}}, u_k; \chi_k)$ evaluated at the encoding context, and then add and subtract $\mathcal{T}_\Theta(z_k^{\text{enc}}, u_k; \chi_{k|s}^{\text{roll}})$ evaluated at the rollout context but at the encoded state:

$$\begin{aligned} e_{k+1|s}^z &= z_{k+1}^{\text{enc}} - \mathcal{T}_\Theta(z_k^{\text{enc}}, u_k; \chi_k) \\ &\quad + \mathcal{T}_\Theta(z_k^{\text{enc}}, u_k; \chi_k) - \mathcal{T}_\Theta(z_k^{\text{enc}}, u_k; \chi_{k|s}^{\text{roll}}) \\ &\quad + \mathcal{T}_\Theta(z_k^{\text{enc}}, u_k; \chi_{k|s}^{\text{roll}}) - z_{k+1|s}^{\text{roll}}. \end{aligned}$$

By the definitions in (17), the first line equals ν_k and the second line equals $\delta_{k|s}^\chi$. The third line can be written as

$$\mathcal{T}_\Theta(z_k^{\text{enc}}, \mathbf{u}_k; \chi_{k|s}^{\text{roll}}) - \mathcal{T}_\Theta(z_{k|s}^{\text{roll}}, \mathbf{u}_k; \chi_{k|s}^{\text{roll}}),$$

where we used the rollout recursion (11). Because $\mathcal{T}_\Theta$ is continuously differentiable in $\mathbf{z}$, the integral mean-value theorem applies:

$$\mathcal{T}_\Theta(z_k^{\text{enc}}, \mathbf{u}_k; \chi_{k|s}^{\text{roll}}) - \mathcal{T}_\Theta(z_{k|s}^{\text{roll}}, \mathbf{u}_k; \chi_{k|s}^{\text{roll}}) = \bar{\Gamma}_{k|s} e_{k|s}^z,$$

with $\bar{\Gamma}_{k|s}$ defined in (19). Combining these steps proves the one-step recursion (18). Repeated substitution from $k = s$ to k yields the finite-horizon expansion (20) with the ordered product $\Phi(k, j)$ as stated.

APPENDIX B PROOF OF THEOREM 2

Differentiate the interpolated transition $\mathcal{T}_\Theta(\mathbf{z}, \mathbf{u}; \chi) = \sum_{i=1}^{m_r} \omega_i(\rho) \mathbf{v}_i(\mathbf{z}, \mathbf{u})$ with respect to $\mathbf{z}$. By the product rule,

$$\Gamma_{\mathbf{z},k} = \frac{\partial \mathcal{T}_\Theta}{\partial \mathbf{z}} = \sum_i \frac{\partial \omega_i}{\partial \mathbf{z}} \mathbf{v}_i + \sum_i \omega_i \frac{\partial \mathbf{v}_i}{\partial \mathbf{z}}.$$

Since $\mathbf{v}_i(\mathbf{z}, \mathbf{u}) = \mathbf{A}_i \mathbf{z} + \mathbf{B}_i \mathbf{u}$, we have $\partial \mathbf{v}_i / \partial \mathbf{z} = \mathbf{A}_i$. The chain rule gives $\partial \omega_i / \partial \mathbf{z} = (\nabla_\rho \omega_i)^\top \partial \rho / \partial \mathbf{z}$. From the scheduling vector definition (6), $\partial \rho / \partial \mathbf{z} = \mathbf{W}_z$. Hence

$$\Gamma_{\mathbf{z},k} = \sum_i \omega_{i,k} \mathbf{A}_i + \sum_i \mathbf{v}_{i,k} (\nabla_\rho \omega_{i,k})^\top \mathbf{W}_z.$$

Substituting (21) into the second term:

$$\sum_i \mathbf{v}_{i,k} (\nabla_\rho \omega_{i,k})^\top \mathbf{W}_z = \sum_i \omega_{i,k} \mathbf{v}_{i,k} (\gamma_{i,k}^\rho - \bar{\gamma}_k^\rho)^\top \mathbf{W}_z.$$

Because $\sum_i \omega_i = 1$, the identity $\sum_i \omega_i (\gamma_i^\rho - \bar{\gamma}^\rho) = \mathbf{0}$ holds. Consequently, we may replace $\mathbf{v}_{i,k}$ by $\mathbf{v}_{i,k} - \bar{\mathbf{v}}_k$ without changing the sum:

$$\sum_i \omega_{i,k} (\mathbf{v}_{i,k} - \bar{\mathbf{v}}_k) (\gamma_{i,k}^\rho - \bar{\gamma}_k^\rho)^\top \mathbf{W}_z = \sum_i \omega_{i,k} \mathbf{v}_{i,k} (\gamma_{i,k}^\rho - \bar{\gamma}_k^\rho)^\top \mathbf{W}_z.$$

This proves the decomposition (22).

For the infinity-norm bound, note that for vectors $\mathbf{a} \in \mathbb{R}^{m_z}$ and $\mathbf{b} \in \mathbb{R}^{m_\rho}$, the induced infinity norm of the rank-one matrix $\mathbf{a}\mathbf{b}^\top$ satisfies $\|\mathbf{a}\mathbf{b}^\top\|_\infty = \|\mathbf{a}\|_\infty \|\mathbf{b}\|_1$. Applying the triangle inequality and submultiplicativity to (22), and using the convexity weights $\sum_i \omega_i = 1$, yields the bound (23).

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